Hash-Based Digital Signatures

Lamport OTS $\rightarrow$ WOTS $\rightarrow$ Merkle Signature Scheme

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Outline

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2. Lamport One-Time Signatures
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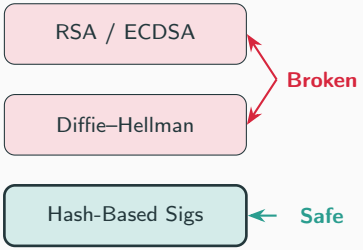
Theory & Implementation

The Quantum Threat

- **Shor's algorithm** (1994) breaks RSA, ECDSA, and Diffie–Hellman in polynomial time on a quantum computer.
- These schemes rely on the hardness of *factoring* or *discrete logs*.
- We need signatures based on different assumptions.



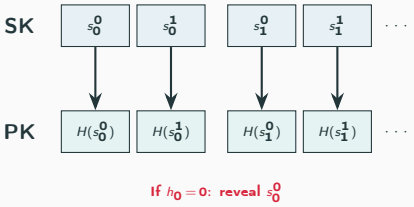
Hash-based signatures rely *only* on the one-wayness of hash functions (e.g. SHA-256).

- No number-theoretic assumptions.
- Grover's algorithm gives only a $\sqrt{}$ speedup $\Rightarrow$ SHA-256 still provides $\sim$128-bit quantum security.

Lamport One-Time Signature

Key Generation: *(Where H is a cryptographic collision resistant hash function e.g. SHA-256)*

- Generate 256 pairs of random 32-byte blocks.
- Secret key: $\mathrm{SK}[i] = (s_i^0,\ s_i^1)$
- Public key: $\mathrm{PK}[i] = \big(H(s_i^0),\ H(s_i^1)\big)$



Signing message m:

1. Compute $h = H(m)$ — 256 bits.
2. For each bit h_i, reveal $s_i^{h_i}$.

Verification:
Check $H(\sigma_i) \stackrel{?}{=} \mathrm{PK}[i][h_i]$ for all i.

Key Properties
Signature = 8,192 B | Signing: 0.03 ms (no hashing!) | Strictly one-time use

Why Is Lamport One-Time?

- Each signature reveals **256 of 512** secret blocks.
- A second signature on a different message reveals *different* blocks.
- After k reuses, an attacker has collected enough blocks to forge *any* message.

Reuses (k)	Blocks Exposed	Forgery Probability
1	50%	$\approx 0\%$
8	99.6%	38.5%
10	99.9%	78%
15	100%	$>99.9\%$

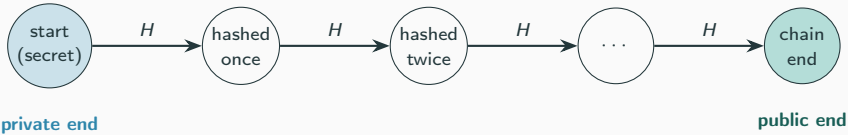
Two solutions: (1) Use WOTS to reduce sizes (2) Use Merkle trees to manage many keys

Winternitz OTS: The Hash Chain

Recall the problem with Lamport: it signs the message *one bit at a time*, so the signature is large (8 KB).

Winternitz idea: sign *several bits at once* as a small number (a "digit"), using a *hash chain*.

A hash chain is one secret value hashed over and over. Each arrow applies H once:

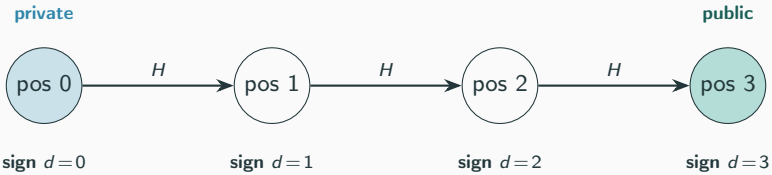


Moving **forward** is easy (just hash). Moving **backward** is infeasible (it would mean inverting H). This one-way direction is what makes the chain secure.

WOTS: Signing One Digit (Toy Example, $w = 4$)

One chain signs **one digit** $d \in \{0, 1, 2, 3\}$. The position along the chain *is* the digit value.

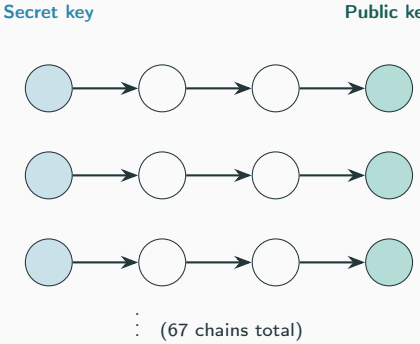
- **Private value:** the start of the chain (random, kept secret).
- **Public value:** the end of the chain — the start hashed $w-1 = 3$ times.
- **To sign digit** d**:** reveal the value at position d (the start hashed d times).
- **To verify:** hash the revealed value the remaining $3 - d$ times — it must reach the public value.



But one chain only signs one digit. A real message hash has many digits — so WOTS uses **many chains in parallel** (next slide).

WOTS: The Full Key is Many Chains

The message hash is split into many digits — and **each digit gets its own chain**. With $w = 16$, that is 67 chains side by side.



Secret key = all 67 chain *starts*.

Public key = all 67 chain *ends*.

Signature = the middle value of each chain, one per digit.

In one sentence

To sign a message: split its hash into digits, and from each chain reveal the value at that digit's position. The verifier finishes every chain and checks all 67 ends match the public key. **(67 = 64 for 256-bit hash + 3 for checksum)**

WOTS: Why We Need a Checksum

The problem: hashing *forward* is free for everyone — including the attacker.

- The attacker can hash a revealed value forward to forge a valid value for any *higher* digit.
- Unprotected, they could forge any message whose digits are all $\geq$ ours.

The fix: a checksum $C = \sum_i (w - 1 - d_i)$, signed with extra chains.

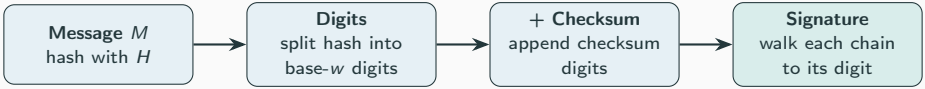
- Raising any message digit makes the checksum *decrease*.
- A smaller checksum digit means going *backward* in a chain — which needs inverting H.
- So the attacker cannot raise a digit without breaking the checksum.

w	Chains (incl. checksum)	Signature Size
4	133	4,256 B
16	**67**	**2,144 B**
256	34	1,088 B

Larger $w \Rightarrow$ fewer chains $\Rightarrow$ smaller signature (but longer chains to compute).

WOTS: The Full Picture

Signing a message is just this pipeline — hash it, split into digits, walk each chain:



The three keys, defined

Secret key — one random start value per chain.

Public key — each start hashed $w-1$ times (every chain end).

Signature — each chain stopped at its digit's position.

Verification

Finish every chain from the signature value; all chain ends must equal the public key.

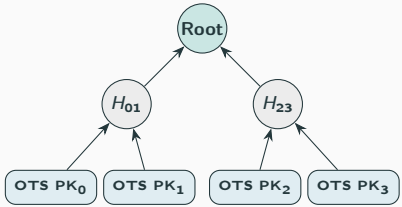
Trade-off: one-time only, but signatures are 4× smaller than Lamport.

From OTS/WOTS to the Merkle Signature Scheme

Both Lamport and WOTS are one-time — a fresh keypair is needed for every message.

Merkle's fix: pre-generate 2^h OTS keypairs and bind them under a single public key using a hash tree.

- Each OTS public key PK_i is hashed to form **leaf** L_i of the tree.
- Internal nodes are computed as $H(\text{left}\|\text{right})$.
- The **tree root** is the single MSS public key (32 B).
- To sign message i: run OTS_i and attach the auth path (h sibling nodes) to let anyone recompute the root.



Lamport or WOTS public keys as leaves

Advantage of WOTS over Lamport
WOTS ($w=16$) signatures are $\approx 4\times$ smaller (2 144 B vs. 8 192 B), directly shrinking the on-wire MSS signature.

Merkle Signature Scheme: Key Generation

Problem: One-Time Signatures (OTS) can only sign a single message securely.

Solution: Build a binary hash tree (Merkle Tree) over 2^h OTS public keys to sign 2^h messages.

Key Generation Process:

1. Generate 2^h independent OTS keypairs.
2. Hash each OTS public key to form the leaves: $\text{Leaf}_i = H(\text{OTS\_PK}_i)$.
3. Compute parent nodes by hashing the concatenation of children: $\text{Parent} = H(\text{left} \| \text{right})$.
4. The **MSS public key** is the tree root (just 32 B).

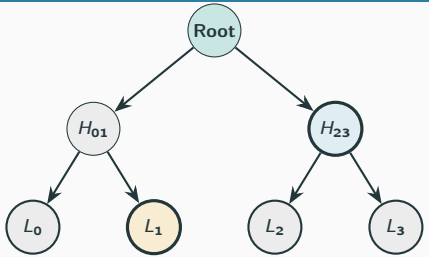
Merkle Signature Scheme: Sign & Verify

To sign the i-th message:

- Sign the message with OTS key i.
- Include the *auth path*: the h sibling hashes needed to reach the root.

To verify:

- Verify the underlying OTS signature against OTS_PK_i.
- Reconstruct the root from Leaf_i and the auth path.
- Check if reconstructed root $\stackrel{?}{=}$ MSS public key.



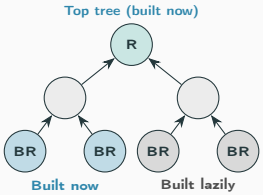
Signing leaf L_1:
Auth path $= [L_0,\ H_{23}]$
Verify: $H(H(L_0 \| L_1) \| H_{23}) = \text{Root}$

Multi-Layer Merkle Trees (Hypertrees)

Problem: Flat MSS KeyGen builds all 2^h OTS keys upfront — exponentially expensive for large h.

Idea: Stack two Merkle trees. The *top tree* is built immediately; each of its leaves certifies the root of a *bottom tree* that is built lazily only when needed.

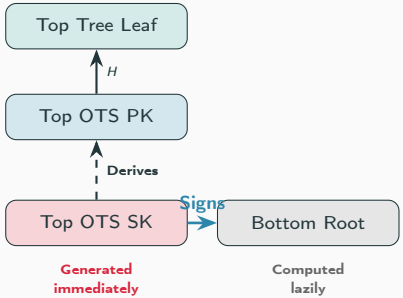
- Total capacity: $N = 2^{h_{\text{top}}+h_{\text{bot}}}$ messages.
- Initial KeyGen: $O(2^{h_{\text{top}}} + 2^{h_{\text{bot}}})$ OTS keys.
- Signing produces *two* MSS signatures: one from the bottom tree, one from the top tree certifying the bottom root.
- This layering is the core idea behind **SPHINCS+ / SLH-DSA** (NIST post-quantum standard).



Hypertrees: The Lazy Link

Question: How is the top tree built if the bottom roots don't exist yet?

1. **Top Tree (Immediate):**
   - Generated from a secret seed.
   - The leaves are just hashes of **Top OTS Public Keys**.
   - Does not contain the bottom root.
2. **Bottom Tree (Lazy):**
   - Built only when signing a message.
   - Hashes up to a single **Bottom Root**.
3. **The Link:**
   - The Bottom Root is **signed** by the corresponding Top OTS Secret Key.



Verification

The verifier checks the signature against the Top OTS PK, hashes it to find the Top Leaf, then climbs the top tree.

Evaluation & Extensions

Evaluation Setup: How Messages Are Generated

Two distinct message generation strategies are used in the evaluation:

Security Simulation (Key-Reuse Attack)

- A mix of **fixed commands** and **random strings**:
  - `"Attack at dawn"`, `"Hold the line"...`
  - `f"Random message {random.random()}"`
- 20 unique messages are signed with the *same* Lamport key.
- A fixed target (`"Retreat at noon"`) tests if a forgery is possible.
- Repeated over **200 independent trials** for statistical validity.

Performance Benchmarking

- A single fixed string is used across all timing trials:

```
"Post-quantum cryptography is exciting"
```

- Each configuration is measured over **5 repetitions**; we report the *mean and variance* to capture both the typical time and its stability.
- The same message is reused across all parameter sweeps for a fair, controlled comparison.

Lamport Key-Reuse Attack

Threat model: A *passive* attacker observes k signatures made with the **same** Lamport key.

- Each signature reveals one block per position (bit 0 or 1).
- After k signatures, the probability a specific block is *still hidden* is $(1/2)^k$.

Forgery probability:

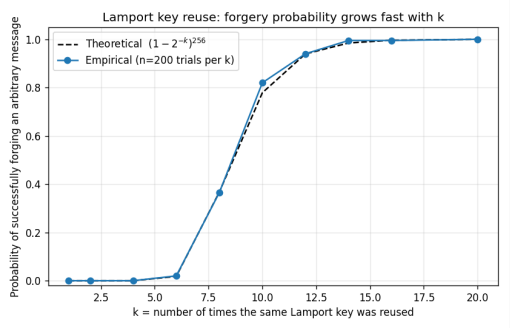
$$P_{\text{forge}}(k) = \left(1 - 2^{-k}\right)^{256}$$

The attacker needs all 256 blocks matching the target message's hash bits. This converges to certainty exponentially.

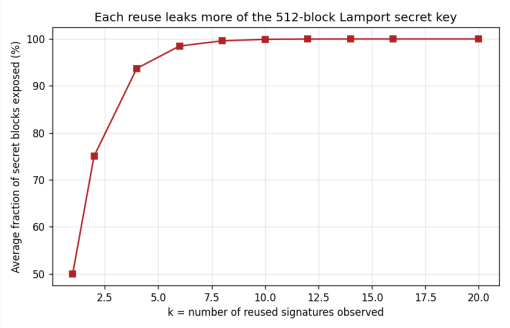
k	P_{forge}	**SK Exposed**
1	$\approx 0\%$	50%
5	0.03%	96.9%
8	38.5%	99.6%
10	78%	99.9%
12	94%	100%
15	99.8%	100%
20	100%	100%

Results from 200 independent trials per k.

Empirical Validation



Empirical vs. theoretical forgery rate.
Theory and experiment match closely.



Fraction of SK exposed vs. reuse count.
By $k = 12$ the complete key is recovered.

OTS Performance Comparison

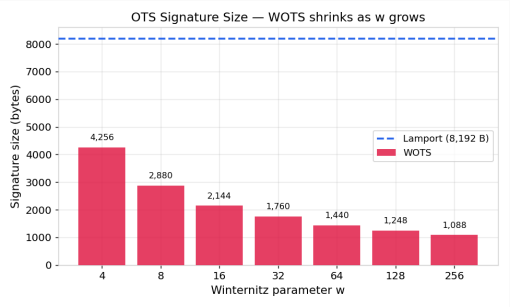
Scheme	KG (ms)	Sign (ms)	Verify (ms)	Sig (B)	PK (B)
Lamport	0.74 ± 0.013	0.02 ± 0.000	0.13 ± 0.002	8,192	16,384
WOTS $w=4$	0.29 ± 0.000	0.11 ± 0.000	0.10 ± 0.000	4,256	4,256
WOTS $w=16$	0.40 ± 0.000	0.19 ± 0.000	0.18 ± 0.000	2,144	2,144
WOTS $w=64$	1.06 ± 0.011	0.43 ± 0.000	0.55 ± 0.000	1,440	1,440
WOTS $w=256$	3.00 ± 0.006	1.56 ± 0.021	1.56 ± 0.014	1,088	1,088

Times reported as **mean** $\pm$ **variance** over 5 runs. The very small variances confirm the measurements are stable.

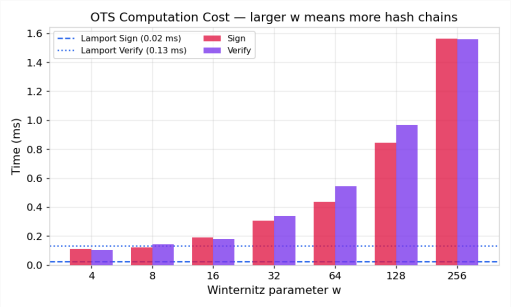
Key observations:

- Lamport signing is extremely fast (0.02 ms) — no hash computation, only copying blocks.
- WOTS $w=256$ is **7.5**$\times$ **smaller** but much slower than Lamport.
- $w=16$ is the practical sweet spot — adopted by XMSS and SPHINCS+.

OTS: Size vs. Speed Trade-off



Signature size decreases *logarithmically* with w.



Computation time grows *linearly* — each chain is $w-1$ hashes long.

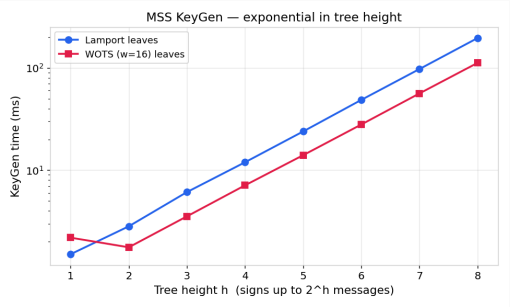
MSS Performance: Scaling with Tree Height

Height h	Messages	KG (ms)	Sign (ms)	Sig Size (B)
Lamport leaves				
4	16	11.96 ± 0.01	0.03 ± 0.00	24,640
6	64	48.73 ± 0.71	0.04 ± 0.00	24,704
8	256	196.28 ± 10.19	0.06 ± 0.00	24,768
WOTS ($w = 16$) leaves				
4	16	7.11 ± 0.03	0.21 ± 0.00	4,420
6	64	27.99 ± 0.17	0.22 ± 0.00	4,484
8	256	112.17 ± 1.13	0.23 ± 0.00	4,548

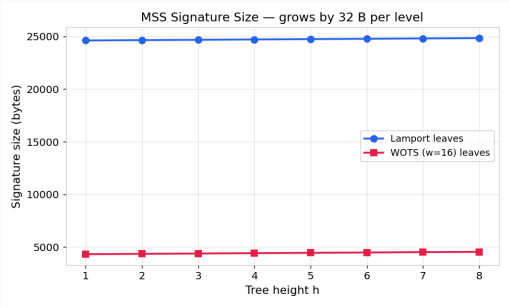
Times reported as **mean** $\pm$ **variance** over 5 runs.

- KeyGen: $O(2^h)$ — doubles with each additional level.
- Sign/Verify: **Constant** regardless of tree height.
- Signature overhead: only **+32 B per level** (one SHA-256 sibling).

MSS: KeyGen is the Bottleneck

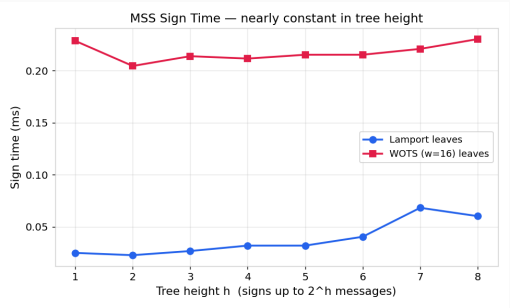


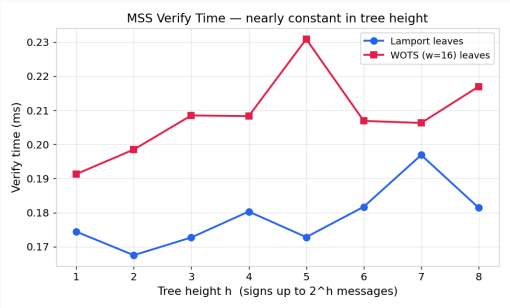
KeyGen scales as $O(2^h)$: the log-scale plot shows a linear trend.



Signature size grows by just 32 B per tree level.

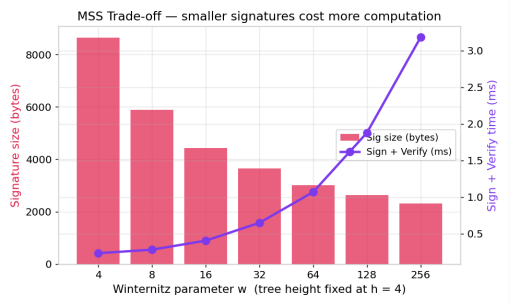
MSS: Sign & Verify are Constant-Time in h





The h additional hashes for the auth path are negligible compared to the underlying OTS operations.

Parameter Tuning: MSS with Varying w



Fixed: $h = 4$ (16 messages).

The *knee* near $w = 16$ marks the optimal balance:

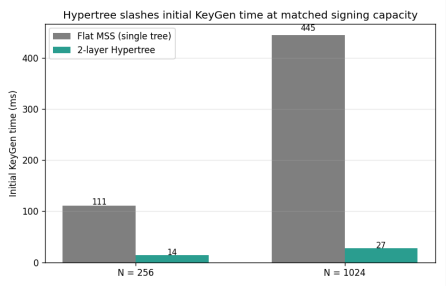
- Below $w = 16$: rapid size improvement with modest time cost.
- Above $w = 16$: diminishing size returns, escalating time cost.

Recommendation
$w = 16$, $h = 4$–8
This matches the XMSS and SPHINCS+ standard configurations.

Multi-Layer Merkle Trees (Hypertrees): Performance

Scheme	N	KG (ms)	Sig (B)
Flat MSS, $h=8$	256	110.7 ± 0.6	4,548
Hypertree, $h_t=h_b=4$	256	13.8 ± 0.01	8,876
Flat MSS, $h=10$	1024	444.9 ± 4.7	4,612
Hypertree, $h_t=h_b=5$	1024	27.4 ± 0.4	8,940

Times: mean $\pm$ variance over 5 runs.



Result: 8.0× faster KeyGen at $N=256$, 16.2× faster at $N=1024$. Signature roughly doubles.

Conclusion

- We built the full pipeline from scratch using SHA-256: Lamport, WOTS, Merkle Signature Scheme, and a two-layer Hypertree.
- WOTS with $w = 16$ is the standard for a reason: four times smaller signatures than Lamport, with manageable speed cost.
- Merkle trees lift the one-time restriction: signing and verifying stay constant, and signatures barely grow with tree height.
- Hypertrees make the scheme practical at large scale: dramatically faster key generation at the cost of roughly doubling the signature size.
- Hash-based signatures rest on only one assumption — that SHA-256 is one-way — which is why they remain secure in the post-quantum era.

Thank You

Questions?

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